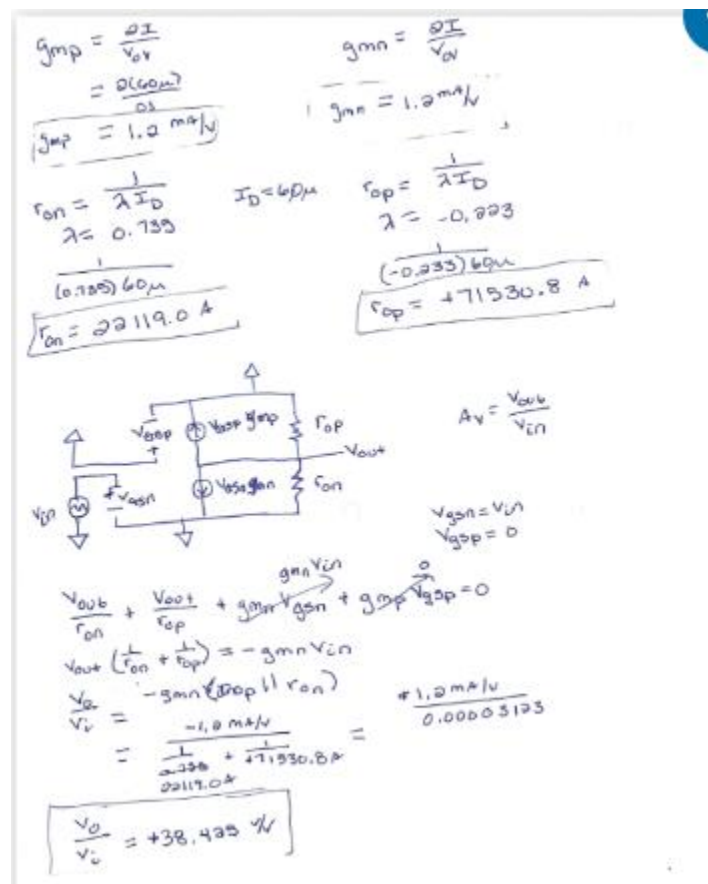


EEE 335 Laboratory 4 – Part 2 Report Submission

Notes (same as Lab 4 – Part 1):

- Do not forget to copyright a curve-based image before you paste it in this report
- Only images that have curves need to be copyrighted.
- Circuit images do not need to be copyrighted if you plan to paste them in this report as long as the title bar has your name as you did in Lab 1
- Scanned documents of your handwritten analysis do not need to be copyrighted
- In this report, only copyright one curve-based image, not more. Choose ANY image in ANY task of your choice to copyright as long as it is an image containing some curve.
- Hand calculations asked in this report can be
 - written on a piece of paper, scanned and pasted, OR
 - typed on a computer, OR
 - provided in ANY other legible method

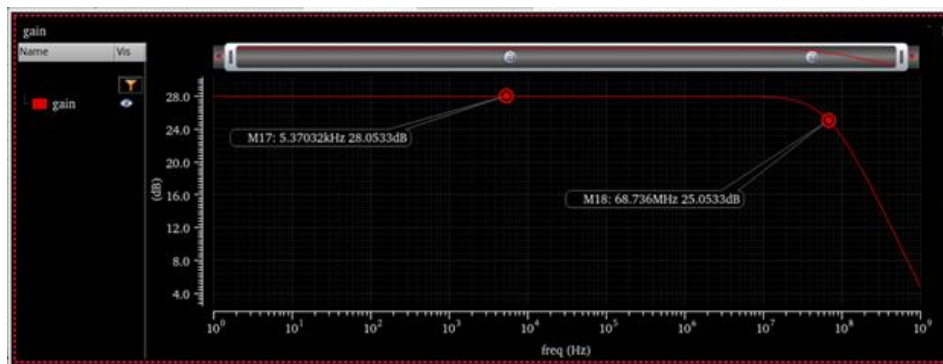
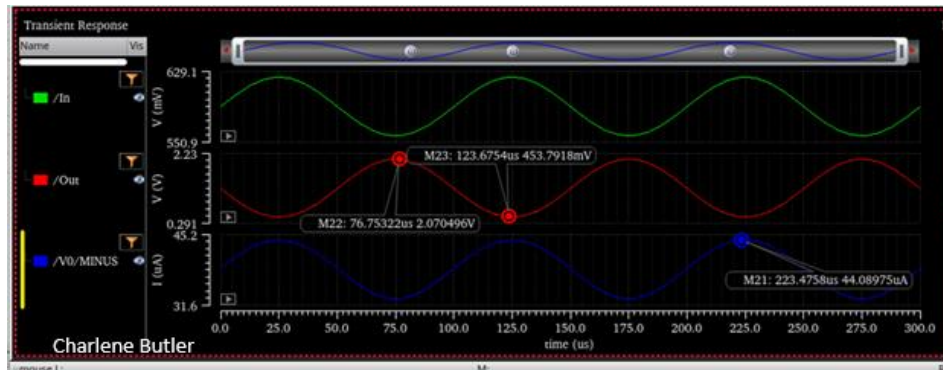
Task 7. Small Signal Analysis (Hand Analysis + Comparison) (5 pts)



Scan 2. Small Signal Analysis

Parameter	Hand Calculation	Simulation Result
ro NMOS	22.12kOhm	31.81kOhm
ro PMOS	71.5kOhm	167kOhm
gm NMOS	1.2mA/V	137.6uA/V
gm PMOS	1.2mA/V	114.2uA/V

Task 8. DC, Transient and AC simulation of the CS Amplifier (12.5 pts)



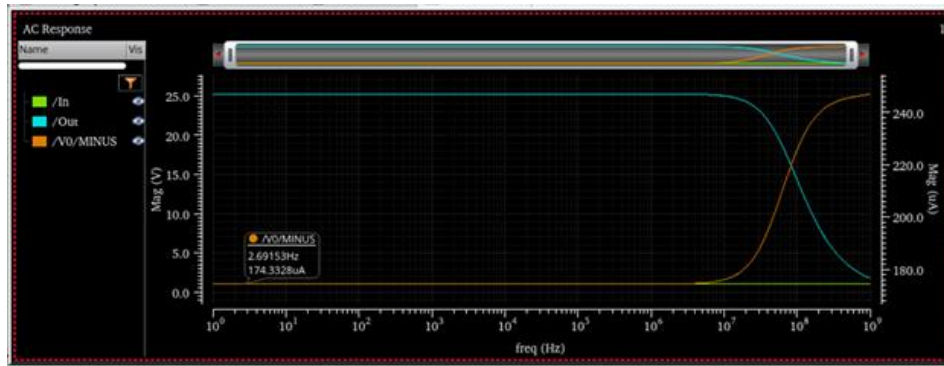


Figure 6. Gain, Transient V_{in} Vs. V_{out} , & AC V_{in} Vs. V_{out}

Task 9. Simulate a V_{out} vs. V_{in} DC transfer curve (VTC) (2.5 pts)



Figure 7. VTC

Task 10. Conclusion (5 pts)

Specification Values	Attained Values
VDD = 2.5 V	VDD = 2.5V
Gain ≥ 20 V/V ≥ 26 dB	Gain = 38.425 V/V = 31.69 dB
I_{Supply} less than 70 μA	$I_{\text{Supply}} = 40\mu\text{A}$
Vpp greater than 2.2 V	Vpp = 1.617V

Table 1. Design Parameters Vs. Attained Values

Conclusion:

If the input voltage is increased a minuscule amount, it will cause the output voltage to drop significantly.

I am not sure why my Vpp is off as I have more than enough gain, the input voltage and current are in the right parameters. I don't know if my graph is maybe off somehow.

I get the use of this lab for sure, I followed all steps as I interpreted them, but I still seem to have errors somehow. I am not sure if it is my math or just an error of some sort in the graph or schematics. I would like to know how this is really supposed to be, maybe a video of how to set everything up and walking through getting the correct answers? I would be very interested in this.